

Exploring FIDE Chess Ratings

An Interactive Visualization of Player Performance, Demographics and Representation

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Abstract

This report documents three interactive visualizations exploring the FIDE chess ratings dataset. The project answers questions across three analytical categories: global insights (Birth Year vs Peak Rating), subset comparison (Gender Rating Gap by Country) and temporal trends (Gender Representation Across Rating Thresholds). Each visualization is designed with interactive features following Shneiderman's Visual Information-Seeking Mantra, enabling users to gain overview, zoom/filter and access details-on-demand.

1 Visualization 1: Birth Year vs Peak Rating

1.1 Research Question

Category: Global Insights

Question: What is the relationship between player age (birth year) and peak rating achievement?

This visualization explores how birth year correlates with peak rating achievement globally, revealing whether there are distinct generational patterns in chess performance.

1.2 Visual Design Overview

The implementation uses an aggregated bubble chart with marginal histograms, enabling overview-to-detail exploration of the dataset.

1.2.1 Visual Encodings

Table 1: Data-to-Visual Mapping

Data Variable	Visual Channel	Rationale
Birth Year	X Position	Position is the strongest channel for precise quantitative comparison
Peak Rating	Y Position	Enables correlation reading against birth year
Player Count (derived)	Bubble Area	Aggregates players into cohorts to prevent overplotting while showing distribution density
Gender	Color Hue	Blue (#4e79a7) for male, red (#e15759) for female, categorical separation
Rating Std Dev (derived)	Opacity	Higher opacity indicates homogeneous cohorts; low opacity indicates varied skill levels within the group

Players are aggregated into configurable birth-year cohorts (default 5 years) and 50-point rating bins to prevent overplotting while preserving patterns. The bubble size is proportional to player count within each (birth bin, rating bin, gender) group.

1.3 Implementation Components

The visualization consists of three coordinated panels:

- **Main Chart:** The central bubble chart showing birth year vs peak rating
- **Top Histogram:** Marginal distribution of birth years (clickable for filtering)
- **Right Histogram:** Marginal distribution of ratings (clickable for filtering)

Additional visual elements include:

- **Trend Lines:** LOESS-smoothed curves per gender showing average rating progression across birth cohorts
- **Density Contours:** Background contour lines indicating concentration zones
- **Ghost Bubbles:** When scrubbing the timeline, dashed ghost circles show players who peaked in the next year

1.4 Implemented Interactions

1.4.1 Interactions Shown in Figures

Figure 1 (Default View): The default state shows all players with bubbles colored by gender and sized by count. The control panel at the top provides access to all interactions: Play/pause button, year readout, Split View toggle, Clear button, cohort size input, timeline slider, country filter and gender legend toggles. The marginal histograms show overall distributions.

Figure 2 (Split View with Country Filter): Split View mode divides the canvas vertically: male players on the left panel, female players on the right, with a central gutter between them. This enables direct visual comparison of rating distributions across genders. The orange/amber colored area in the gutter represents the rating gap between genders at each birth cohort. Additionally, the country filter dropdown has been used to show data for a specific federation only, demonstrating how the visualization supports country-wise analysis (one country at a time).

Figure 3 (Histogram Filtering): Clicking on histogram bars filters the main chart to show only those cohorts. Both birth year and rating cohorts can be selected (multi-select supported). The histogram bar turns purple with a pulsing effect to indicate selection. This allows detailed comparison between specific age groups or rating tiers.

1.4.2 Additional Interactions (Not Shown)

- **Timeline Animation/Scrubbing:** The Play button animates through years (800ms per step), progressively showing players who achieved their peak by each year. The slider can be manually dragged for instant time navigation. Ghost bubbles (dashed circles) preview players who will peak in the next year.
- **Bubble Hover:** Hovering a bubble displays a tooltip with cohort statistics (player count, average rating, standard deviation, top 3 countries) and draws dashed connector lines to corresponding histogram bars.
- **Gender Toggle:** Clicking legend items (Male/Female) shows/hides that gender's bubbles and trend lines. When one gender is hidden, the Y-axis rescales to fit the remaining data.
- **Axis Zoom:** Double-clicking on the chart zooms to that region (4× scale). A “Reset Zoom” button appears to restore the original view.

- **Configurable Cohort Size:** Users can adjust the birth cohort bin size (1–50 years) via an input field, dynamically re-aggregating the data.

1.5 Screenshots

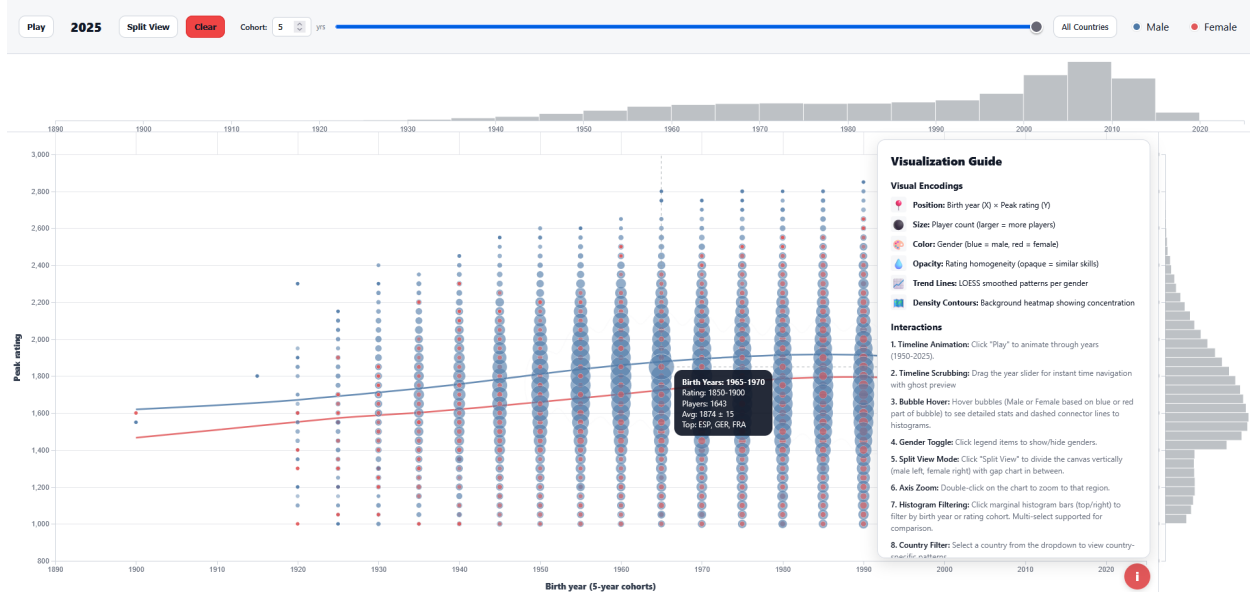


Figure 1: Default view showing the aggregated bubble chart with all players. Bubbles are colored by gender (blue=male, red=female) and sized by player count. Marginal histograms show birth year (top) and rating (right) distributions. LOESS trend lines indicate average rating progression per gender.

1.6 How the Visualization Answers the Question

The visualization addresses the research question through multiple mechanisms:

1. **Overview Pattern:** The bubble positions immediately reveal that peak ratings have risen over successive generations. Players born in later decades achieve higher ratings on average.
2. **Distribution Shape:** Bubble sizes show where most players cluster. The visualization reveals that the majority of rated players were born in the 1980s–2000s.
3. **Trend Analysis:** LOESS-smoothed trend lines separate signal from noise, showing the overall upward trajectory across birth cohorts.
4. **Temporal Exploration:** Timeline scrubbing allows users to observe how the rating landscape evolved over time, answering “what did the age-rating distribution look like at different points in history?”
5. **Cohort Comparison:** Histogram filtering enables focused analysis of specific age groups, comparing their rating distributions.

1.7 Visual Information-Seeking Mantra

This visualization implements Shneiderman’s (1996) three-tier interaction model:

Overview First: The aggregated bubble chart displays the complete birth year vs peak rating landscape at a glance. All cohorts are visible simultaneously with marginal histograms providing distributional context. Bubble sizes encode player counts, enabling immediate identification of where most players cluster. LOESS trend lines summarize overall patterns without obscuring individual data points.

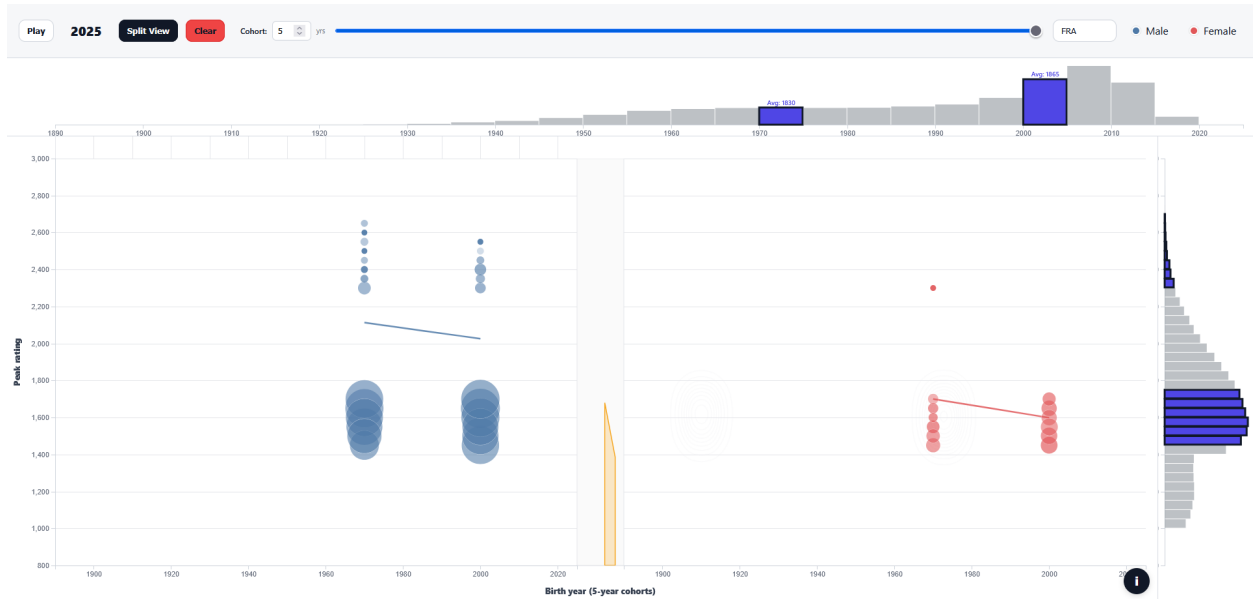


Figure 2: Split View mode with country filter applied. The canvas divides into male (left) and female (right) panels, showing only players from a specific federation. The central amber area shows the rating gap between genders. This demonstrates how the visualization supports country-wise comparison.

Zoom and Filter:

- Temporal filtering via timeline slider and animation (Gapminder-style scrubbing)
- Birth cohort filtering via clickable histogram bars (multi-select supported)
- Rating cohort filtering via right histogram bars
- Gender filtering via legend toggle (show/hide male or female)
- Country filtering via dropdown selector
- Axis zoom via double-click with smooth animated transitions
- Split View mode for gender-separated analysis

Details-on-Demand:

- Bubble hover reveals cohort statistics (player count, average rating, standard deviation, top countries)
- Dashed connector lines link hovered bubble to corresponding histogram bars
- Configurable cohort size enables granularity adjustment
- Selection stats panel shows aggregate information for filtered subsets

1.8 Deviations from Initial Design

Justification: The histogram-based filtering is more discoverable and intuitive than rectangular brush selection, which can conflict with zoom interactions. Configurable cohort size was added to support different levels of granularity based on user needs.

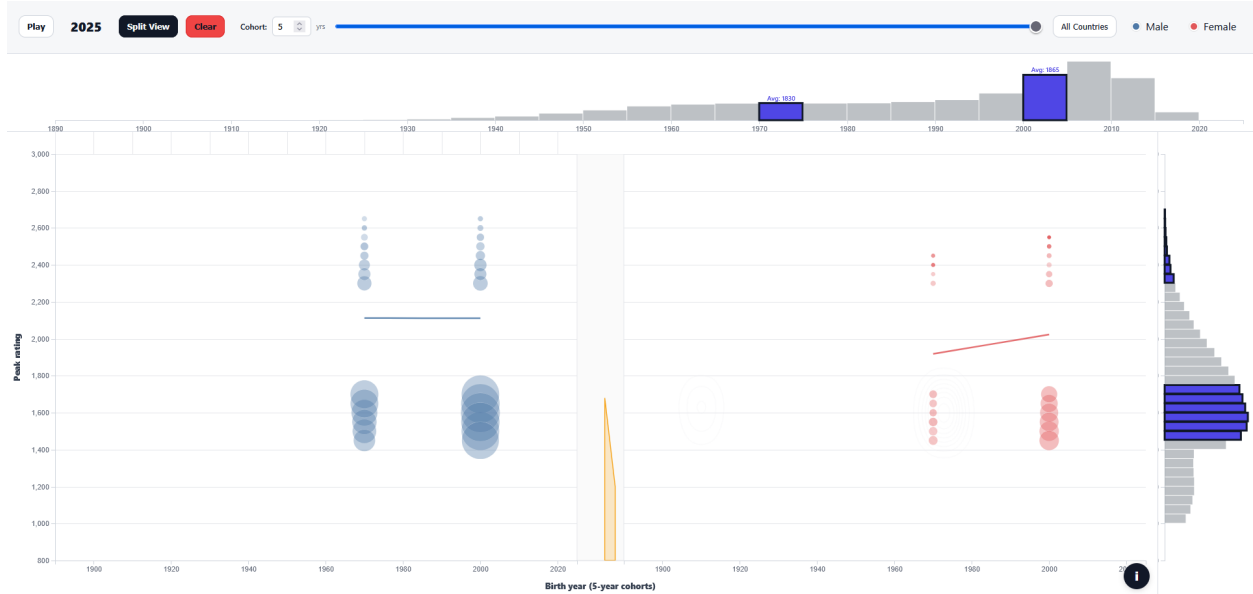


Figure 3: Histogram filtering: clicking on birth year or rating histogram bars filters the main chart to show only those cohorts, enabling focused comparison of specific age groups or rating tiers.

Table 2: Differences from Initial Design

Feature	Initial Design	Implementation
Brushing Selection	Click-drag rectangle with detail panel sliding in	Replaced with histogram-based multi-select filtering (more intuitive)
Timeline Range	Fixed 1950–2025	Dynamically computed from actual data range
Cohort Size	Fixed 5-year bins	User-configurable (1–50 years) via input field
Histogram Bars	Single-select only	Multi-select supported for comparison

2 Visualization 2: Gender Rating Gap by Country

2.1 Research Question

Category: Comparing Subsets

Question: How does the rating gap between genders vary across countries and has it changed over time?

This visualization explores the multi-dimensional comparison of gender, geography and temporal dimensions to reveal where gender parity is improving or declining across chess federations.

2.2 Visual Design Overview

The implementation provides three distinct view modes for exploring the same data from different perspectives: a geographic map, a ranked bar chart and a continental hexagon grid.

2.2.1 Visual Encodings

The gender gap is computed as $Mean(Rating_{Male}) - Mean(Rating_{Female})$ per country per year. Countries with fewer than 3 players of each gender are excluded.

Table 3: Data-to-Visual Mapping

Data Variable	Visual Channel	Rationale
Country	Position (Geographic)	Map view preserves spatial intuition for regional pattern recognition
Gender Gap	Color (Diverging)	Blue for male advantage, red for female advantage, white for parity, diverging scale shows direction and magnitude
Gap Magnitude	Color Saturation	More saturated colors indicate larger gaps; near-white indicates parity
Time Year	Animation/Slider	Reveals temporal evolution of gaps across countries
Continent	Spatial Grouping	Groups countries in Continental view for regional comparison

2.3 Implementation Components

The visualization supports three view modes:

- **Map View:** Choropleth world map with countries colored by gender gap. Supports zoom and pan for detailed exploration.
- **Ranked View:** Horizontal bar chart with all countries sorted by gap magnitude. Bars extend left (female advantage) or right (male advantage) from a central zero line.
- **Continental View:** Hexagonal tiles grouped by continent, reducing geographic clutter while preserving categorical grouping.

Side panel components:

- **Selected Countries Panel:** Shows comparison of clicked countries with sparkline trend charts (2015–2025)
- **Statistics Panel:** Displays global or filtered statistics (average gap, male/female averages, country count)
- **Legend:** Diverging color scale from Male +300 to Female +300

2.4 Implemented Interactions

2.4.1 Interactions Shown in Figures

Figure 4 (Map View with Selection and Hover): The default Map View shows a choropleth world map where countries are colored by their gender rating gap. Blue indicates male advantage (males have higher average ratings), while red would indicate female advantage. A country is selected (shown with orange border) and its details appear in the side panel. The hover tooltip shows country name, gap value and player counts. The Selected Countries panel displays a sparkline showing the gap trend from 2015–2025.

Figure 5 (Continental View with Tiles): Continental View arranges countries as hexagonal tiles grouped by continent (Europe, Asia, Americas, Africa, Oceania). Each tile is colored by gap value. This view is particularly useful when you want to compare countries within a continent or across continents without geographic distortion. Selected countries are highlighted with orange borders.

Figure 6 (Ranked View with Europe Filter): Ranked View displays countries as horizontal bars sorted by gap magnitude. A continent filter (Europe) has been applied, showing only European countries. Bars extend from the center: right for male advantage, left for female advantage (if any). The sort toggle allows switching between “Male Advantage First” and “Female Advantage First” ordering. This view is optimal for comparing exact gap magnitudes across countries.

2.4.2 Additional Interactions (Not Shown)

- **Timeline Animation:** The Play button animates through years 2015–2025, with colors smoothly transitioning to show how gaps evolve over time.
- **Multi-Country Comparison:** Clicking multiple countries adds them to the comparison panel, each showing a sparkline of their gap history.
- **Map Zoom/Pan:** In Map View, users can zoom (scroll) and pan (drag) to explore regions in detail.
- **Clear Selection:** A “Clear All” button removes all selected countries from the comparison panel.
- **Info Panel:** An info button (i) toggles a guide panel explaining view modes, interactions and color encoding.

2.5 Screenshots

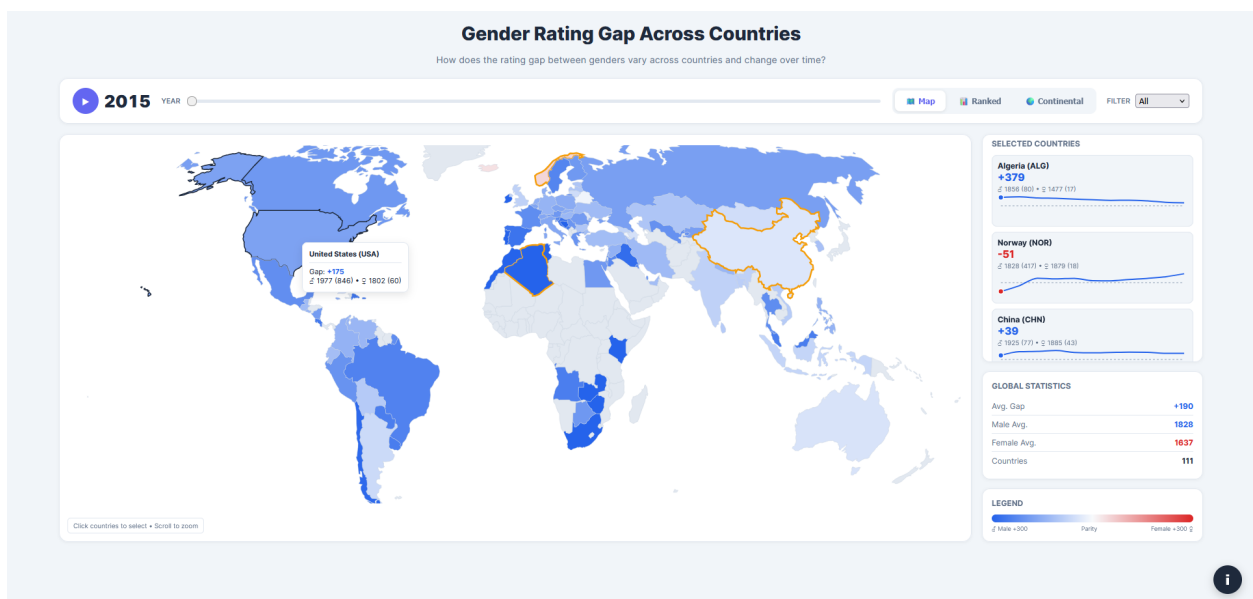


Figure 4: Map View showing the choropleth world map with countries colored by gender rating gap. A country is selected (orange border) with details shown in the side panel including a sparkline trend. The tooltip displays gap details on hover.

2.6 How the Visualization Answers the Question

The visualization addresses the research question through multiple mechanisms:

1. **Geographic Patterns:** Map View immediately reveals regional patterns, whether certain geographic areas have larger or smaller gender gaps.
2. **Precise Ranking:** Ranked View allows exact comparison of gap magnitudes, identifying which countries have the largest gaps and which approach parity.
3. **Continental Comparison:** Continental View groups countries for easy within-region and cross-region comparison without geographic distortion.
4. **Temporal Evolution:** The timeline slider and sparklines answer “has it changed over time?” by showing how each country’s gap has evolved from 2015 to 2025.

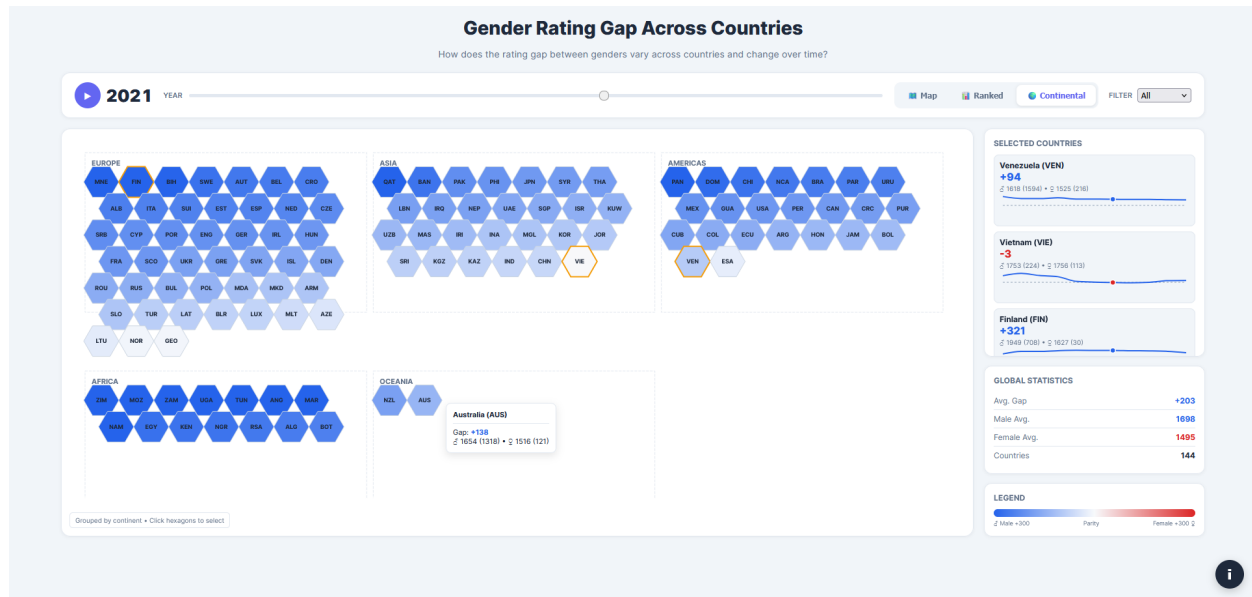


Figure 5: Continental View with hexagonal tiles grouped by continent. Each tile represents a country, colored by its gender gap. This view reduces geographic clutter and enables easy within-continent and cross-continent comparison.

5. **Multi-Country Comparison:** Selecting multiple countries enables direct side-by-side comparison of their gap trends.

2.7 Visual Information-Seeking Mantra

This visualization implements Shneiderman’s (1996) three-tier interaction model:

Overview First: The Map View presents a global overview of gender gaps across all countries simultaneously. The choropleth coloring enables rapid identification of regional patterns through pre-attentive color perception. The Ranked View provides an ordered overview of all countries by gap magnitude, while Continental View offers a structured overview grouped by region.

Zoom and Filter:

- Temporal filtering via timeline slider and Play animation (2015–2025)
- Continent filtering via dropdown selector
- View mode switching between Map, Ranked and Continental layouts
- Map zoom and pan for geographic exploration
- Sort order toggle in Ranked View (Male Advantage First / Female Advantage First)

Details-on-Demand:

- Hover tooltips reveal country-specific details (gap value, player counts, averages)
- Click-to-select adds countries to the comparison panel with sparkline trend charts
- Side panel statistics dynamically update based on active continent filter
- Multi-country selection enables direct comparison of gap evolution

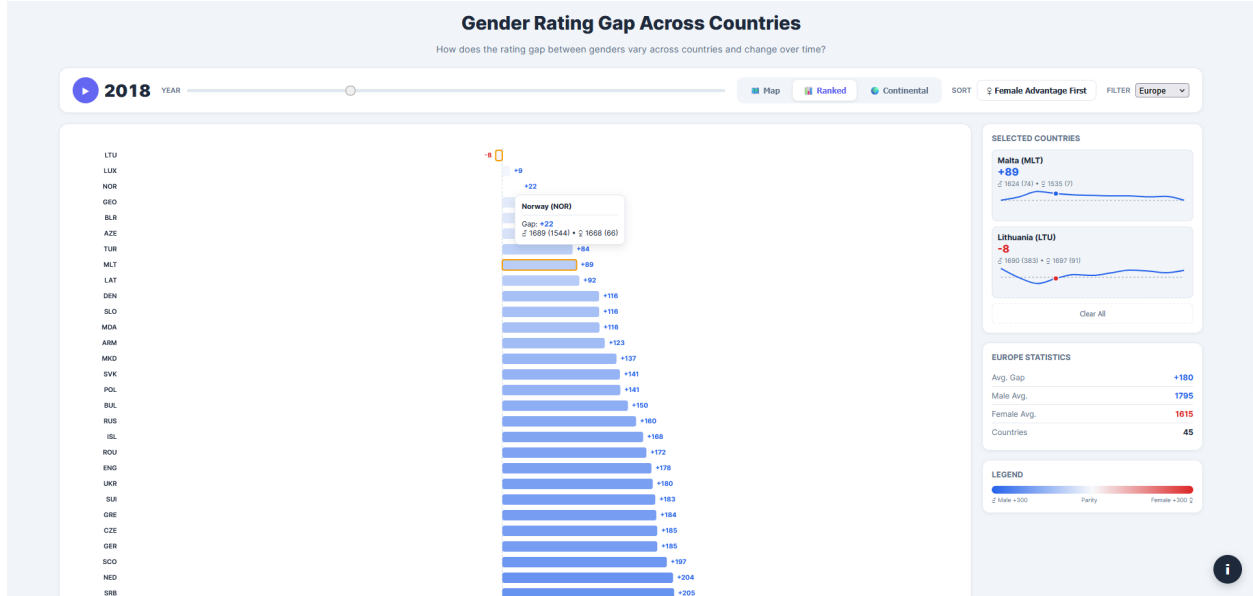


Figure 6: Ranked View showing European countries as horizontal bars sorted by gap magnitude. The continent filter is set to Europe. Bars extend right for male advantage. This view enables precise comparison of gap magnitudes across countries.

Table 4: Differences from Initial Design

Feature	Initial Design	Implementation
Tile Layout	Hexagonal tiles approximating geographic positions	Hexagons only in Continental View; Map View uses actual choropleth
Sparkline in Tiles	Embedded within each tile	Sparklines shown in side panel for selected countries only
Player Total / Sample Size Encoding	Saturation and border thickness	Not implemented, simplified to focus on gap value
Search and Highlight	Search box with gold glow on matches	Not implemented

2.8 Deviations from Initial Design

Justification: The choropleth map provides more intuitive geographic recognition than abstracted hex tiles. Moving sparklines to the side panel avoids visual clutter and allows larger, more readable trend lines. Sample size encoding was omitted to simplify the visual, countries with insufficient data are simply excluded.

3 Visualization 3: Gender Representation Timeline

3.1 Research Question

Category: Understanding Distributions and Trends

Question: How has gender representation evolved across different rating thresholds over time?

This visualization tracks the temporal evolution of gender composition within distinct skill tiers (Beginners, Intermediate, Advanced, Expert, Elite) from 2015–2025, revealing whether participation patterns differ across performance levels.

3.2 Design Rationale

While gender representation could potentially be integrated into Visualization 1 or 2, doing so would create visual clutter and dilute the focus of those visualizations. Visualization 1 focuses on the age-rating relationship with aggregated cohorts, while Visualization 2 emphasizes geographic comparison of gender gaps. This third visualization is deliberately separated to provide a dedicated, uncluttered view of temporal representation trends across skill levels, a fundamentally different analytical question.

3.3 Visual Design Overview

The implementation uses small multiples of stacked area charts, one panel per rating threshold, enabling simultaneous comparison across skill levels while preserving temporal resolution.

3.3.1 Visual Encodings

Table 5: Data-to-Visual Mapping

Data Variable	Visual Channel	Rationale
Time (Month)	X Position	Horizontal time axis for temporal pattern recognition
Player Count	Y Position (Area Height)	Stacked areas show both total and gender composition
Gender	Color Hue	Blue for male, red for female: consistent with other visualizations
Rating Threshold	Facet (Small Multiple)	Separate panels per threshold enable cross-tier comparison via Gestalt proximity

The stacked area design reveals both absolute counts (total height) and relative proportions (area ratios) simultaneously. Female players are placed at the bottom to make their growth more visually trackable.

3.4 Implementation Components

Small Multiples Layout: Five panels arranged in a 3×2 grid, each representing a rating threshold:

- < 1500 (Beginners)
- 1500–2000 (Intermediate)
- 2000–2400 (Advanced)
- 2400–2600 (Expert)
- 2600+ (Elite)

Summary Statistics Panel: Displays aggregate statistics for currently visible categories:

- Total Players
- Male Players (count)
- Female Players (count and percentage)
- Most Popular threshold

Theme Support: Light and dark mode toggle for user preference and accessibility.

3.5 Implemented Interactions

3.5.1 Interactions Shown in Figures

Figure 7 (Default View): The default state shows all five rating threshold panels as stacked area charts spanning 2015–2025. The timeline is set to the latest month, showing the complete temporal evolution. Blue areas represent male players, red areas represent female players. The current timeline position is indicated by a vertical yellow line. Summary statistics at the bottom show totals across all visible categories.

Figure 8 (Percentage Mode with Category Filtering): This screenshot demonstrates multiple interactions: (1) Dark mode theme is active, (2) View mode is switched to Percentage (showing gender proportions instead of absolute counts) and (3) Two categories have been hidden using the category filter. The summary statistics update to reflect only the visible categories. The “Hiding” banner indicates which categories are filtered out.

3.5.2 Additional Interactions (Not Shown)

- **Timeline Animation:** Play button animates through months with configurable speed ($0.5\times$, $1\times$, $2\times$, $5\times$). The yellow position line moves across panels as time progresses.
- **Timeline Scrubbing:** Drag the slider to jump to any month. Keyboard shortcuts ($\leftarrow$ $\rightarrow$) allow frame-by-frame navigation.
- **Hover Tooltips:** Hovering on any chart shows a dashed vertical line and tooltip with exact counts for that month/threshold (total, male, female, percentages).
- **Gender Toggle:** Clicking legend items (Male/Female) hides that gender’s area, enabling focus on one gender’s trend.
- **Category Mode:** Dropdown switches between “Hide Selected” and “Show Only Selected” for flexible filtering.
- **Keyboard Shortcuts:** Space to play/pause, arrow keys to scrub timeline.

3.6 Screenshots

3.7 How the Visualization Answers the Question

The visualization addresses the research question through multiple mechanisms:

1. **Temporal Trends:** The area chart format immediately reveals whether female representation is increasing, stable, or declining over time at each skill level.
2. **Cross-Threshold Comparison:** Small multiples enable direct comparison, users can see that female representation is higher at beginner levels and nearly zero at elite levels.
3. **Absolute vs Proportional:** Toggle between count and percentage views answers both “how many?” and “what proportion?”
4. **Focus Filtering:** Category filtering allows focused analysis on specific skill tiers of interest.
5. **Animation:** Temporal animation reveals the dynamics of change, whether growth is steady, accelerating, or stagnating.

3.8 Visual Information-Seeking Mantra

This visualization implements Shneiderman’s (1996) three-tier interaction model:

Overview First: The small multiples layout displays all five rating threshold categories simultaneously, with the complete temporal range visible (2015–2025). Each panel shows the full gender distribution evolution as stacked areas, enabling rapid cross-threshold comparison through Gestalt proximity principles.

Zoom and Filter:

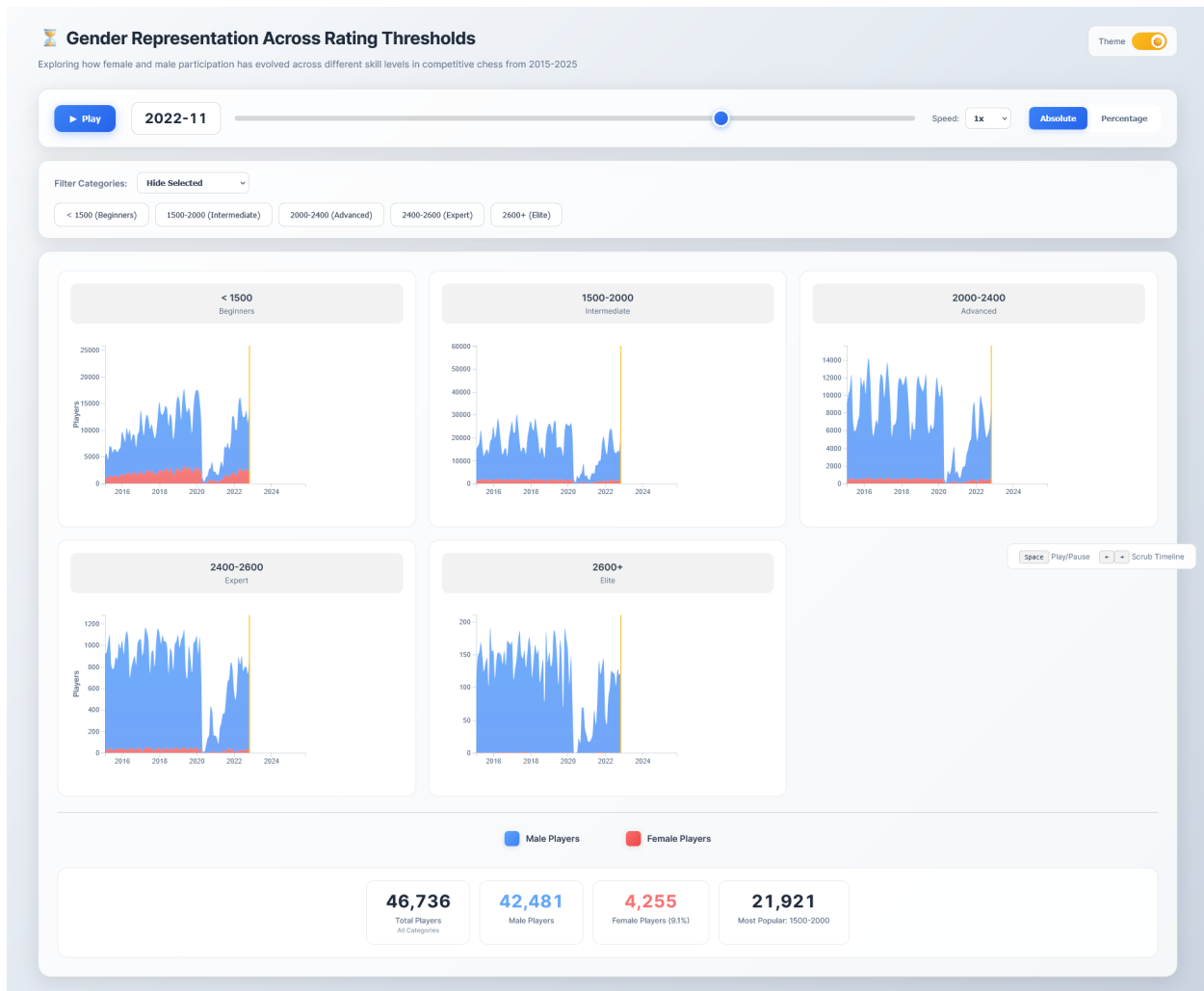


Figure 7: Default view: Small multiples of stacked area charts for each rating threshold. Blue represents male players, red represents female players. The timeline slider and summary statistics are visible. All five categories are shown with the full 2015–2025 range.

- Temporal filtering via animated timeline scrubber (Gapminder-style)
- Category selection via click-to-toggle mechanism with “Hide Selected” vs “Show Only Selected” modes
- View mode switching between absolute counts and percentage distributions
- Gender filtering via legend interaction

Details-on-Demand:

- Hover interaction reveals temporal cross-section with exact counts and percentages
- Visual feedback through hover lines and tooltips
- Summary statistics dynamically update based on active category selections

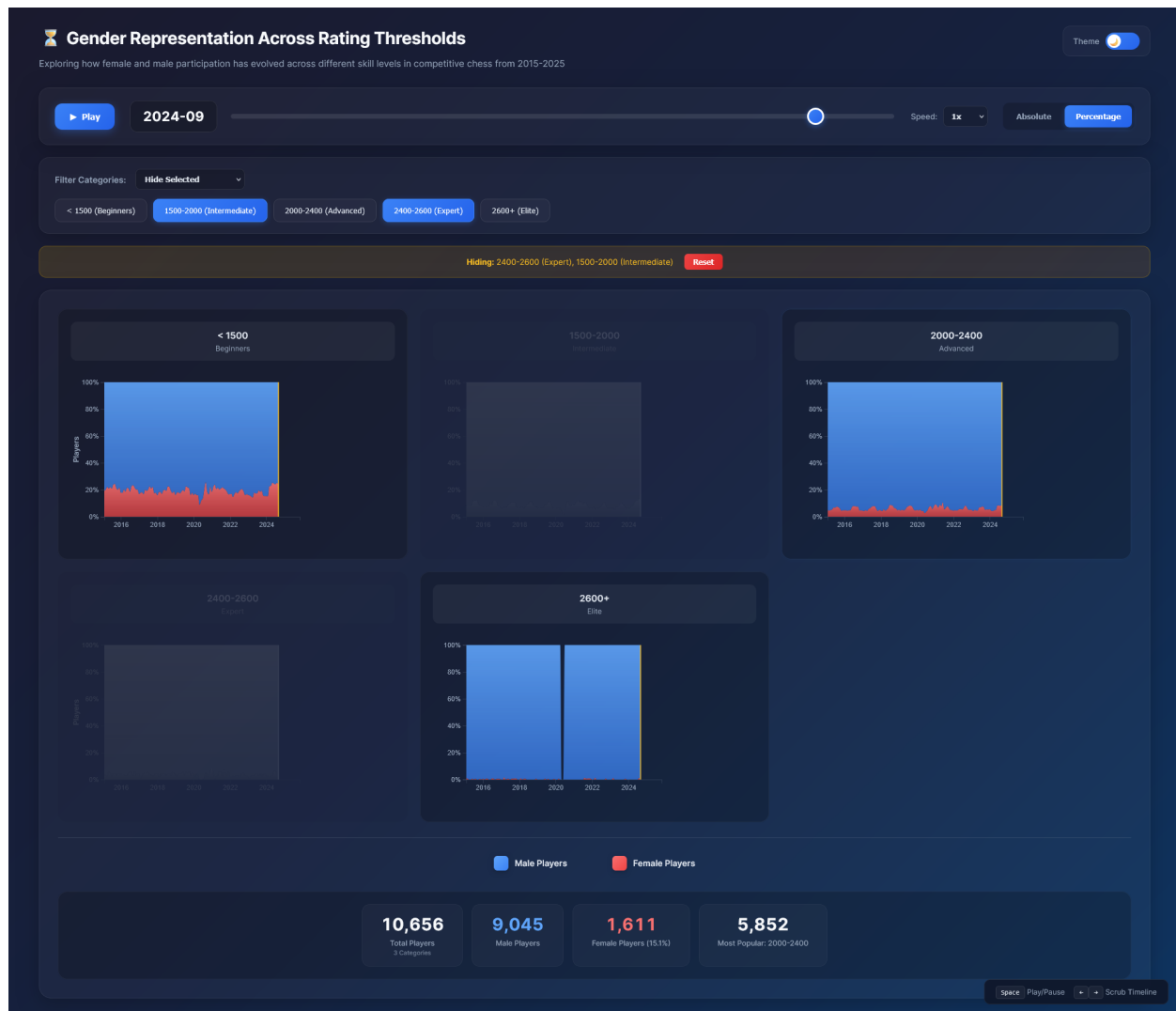


Figure 8: Filtered view in dark mode with percentage display. Two rating categories have been hidden (shown in the yellow banner). The Y-axis now shows 0–100% instead of absolute counts, revealing gender proportion trends more clearly.